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NEET · CHEMISTRY · CHAPTER-WISE PYQS

Haloalkanes and Haloarenes

NEET Chemistry · Previous Year Questions with Answer Key

52	35	1988 – 2025	1	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (More than One Correct Answer)	1 question
MCQ (Single Correct Answer)	50 questions
Out of syllabus	1 question
Answer key	at the end

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MCQ (More than One Correct Answer) 1 questions

Q1. NEET 2025

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) :



undergoes S_N2 reaction faster than



Reason (R) : Iodine is a better leaving group because of its large size. In the light of the above statements, choose the correct answer from the options given below:

(A) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

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MCQ (Single Correct Answer) 50 questions

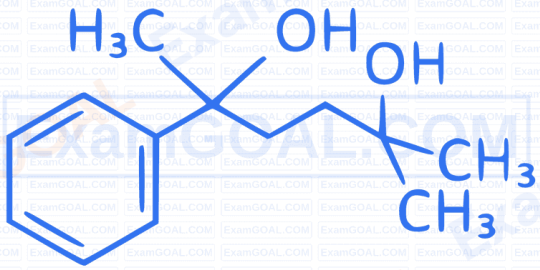
Q2. NEET 2025

Open solution

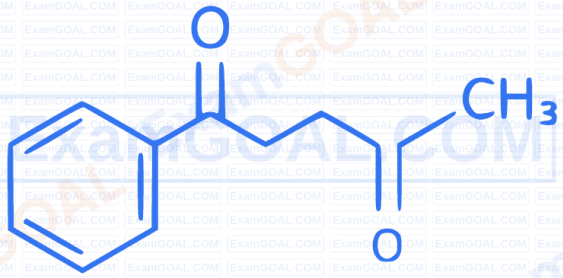
The major product of the following reaction is



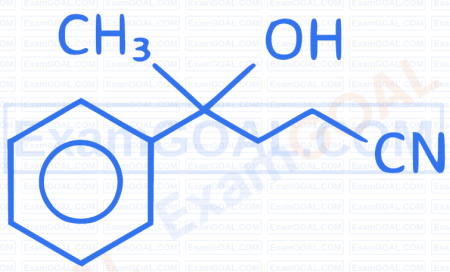
(A)



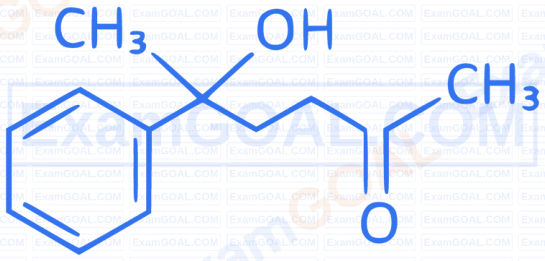
(B)



(C)



(D)



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Q3. NEET 2024 (Re-Examination)

Open solution

The major product C in the below mentioned reaction is:

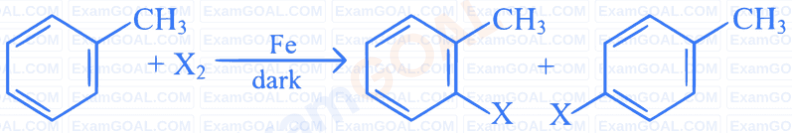
$$CH_3CH_2CH_2Br \xrightarrow[\Delta]{\text{alc. KOH}} A \xrightarrow{HBr} B \xrightarrow[\Delta]{\text{aq. KOH}} C$$

(A) Propan-1-ol (B) Propan-2-ol (C) Propane (D) Propyne

Q4. NEET 2024 (Re-Examination)

Open solution

The following reaction method



is not suitable for the preparation of the corresponding haloarene products, due to high reactivity of halogen, when X is :

(A) F (B) I (C) Cl (D) Br

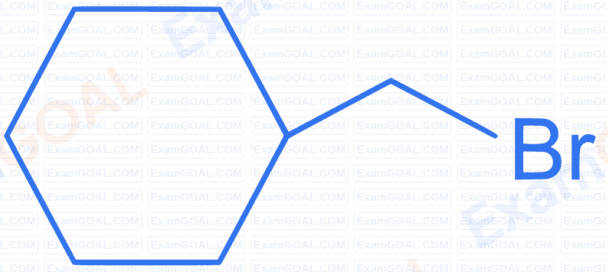
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Q5. NEET 2024

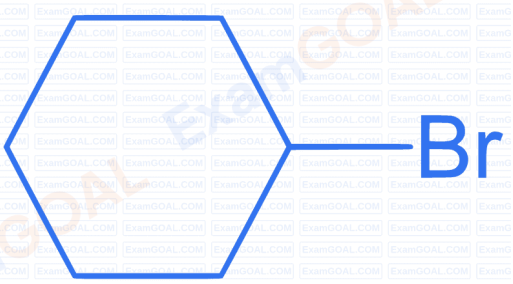
The compound that will undergo S_N1 reaction with the fastest rate is

Open solution

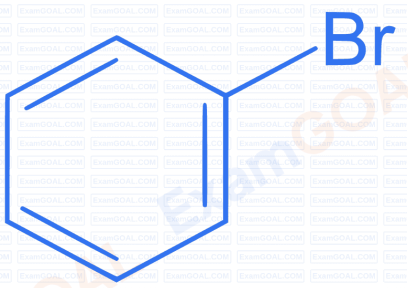
(A)



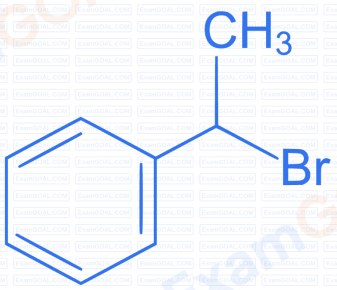
(B)



(C)



(D)



Q6. NEET 2024

Identify the major product C formed in the following reaction sequence :

$$CH_3 - CH_2 - CH_2 - I \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow[Br_2]{NaOH} \underset{(\text{major})}{C}$$

Open solution

(A) propylamine

(B) butylamine

(C) butanamide

(D) α-bromobutanoic acid

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Q7. NEET 2023 Manipur

Open solution

Which amongst the following reactions of alkyl halides produces isonitrile as a major product?

(A) $\mathrm{R-X+HCN\rightarrow}$

(B) $\mathrm{R-X+AgCN\rightarrow}$

(C) $\mathrm{R-X+KCN\rightarrow}$

(D) $\mathrm{R-X+NaCN\xrightarrow[C_2H_5OH]{H_2O}}$

Choose the most appropriate answer from the options given below:

(A) (D) only **(B)** (C) and (D) only **(C)** (B) only **(D)** (A) and (B) only

Q8. NEET 2023 Manipur

Open solution

Choose the correct sequence of reagents in the conversion of 4-nitrotoluene to 2-bromotoluene.

(A) $\mathrm{NaNO_2/HCl;Sn/HCl;Br_2;H_2O/H_3PO_2}$

(B) $\mathrm{Sn/HCl;NaNO_2/HCl;Br_2;H_2O/H_3PO_2}$

(C) $\mathrm{Br_2;Sn/HCl;NaNO_2/HCl;H_2O/H_3PO_2}$

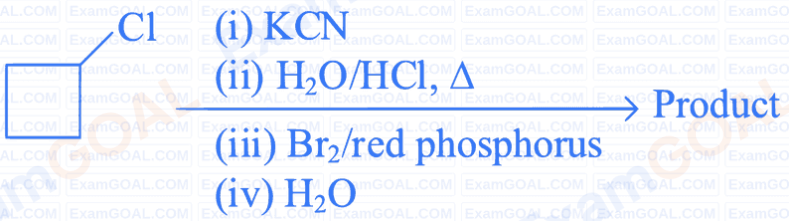
(D) $\mathrm{Sn/HCl;Br_2;NaNO_2/HCl;H_2O/H_3PO_2}$

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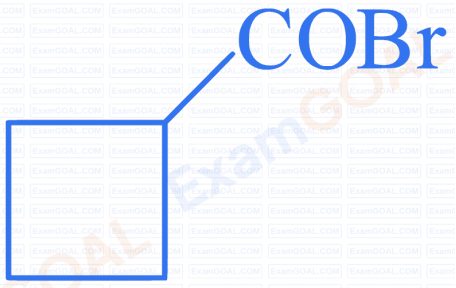
Q9. NEET 2023 Manipur

Open solution

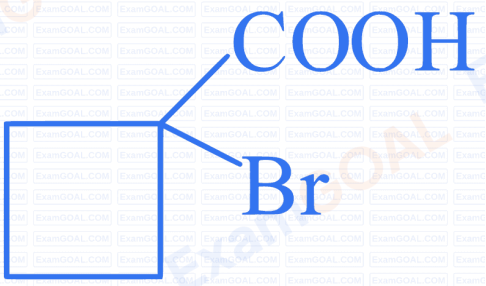
Identify the product in the following reaction



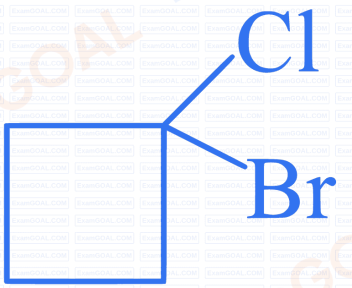
(A)



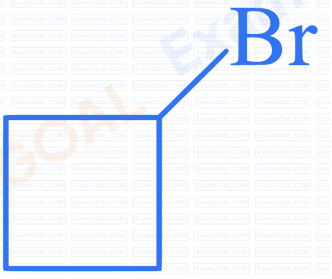
(B)



(C)



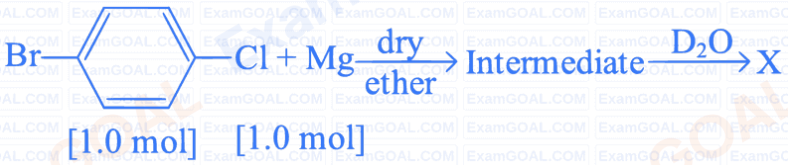
(D)



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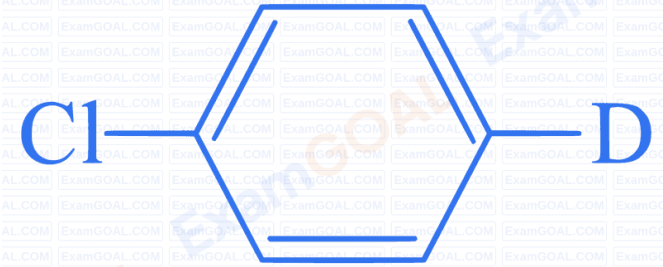
Q10. NEET 2023 Manipur

Identify 'X' in the following reaction.



Open solution

(A)



(B)



(C)



(D)

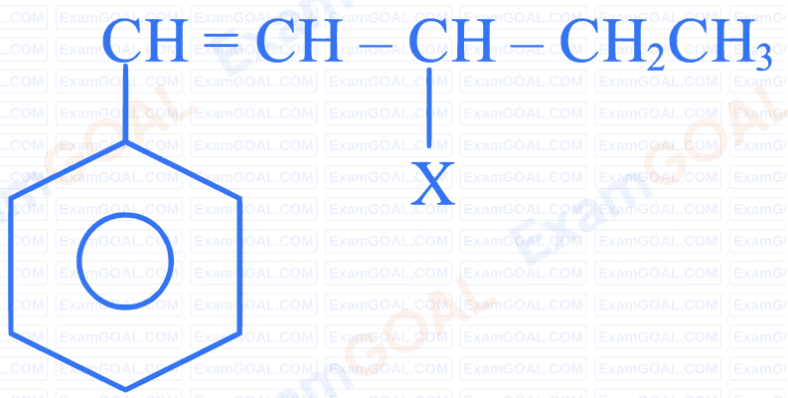


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Q11. NEET 2023

Open solution

The given compound



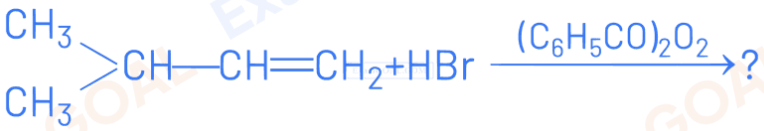
is an example of _____.

(A) aryl halide (B) allylic halide (C) vinylic halide (D) benzylic halide

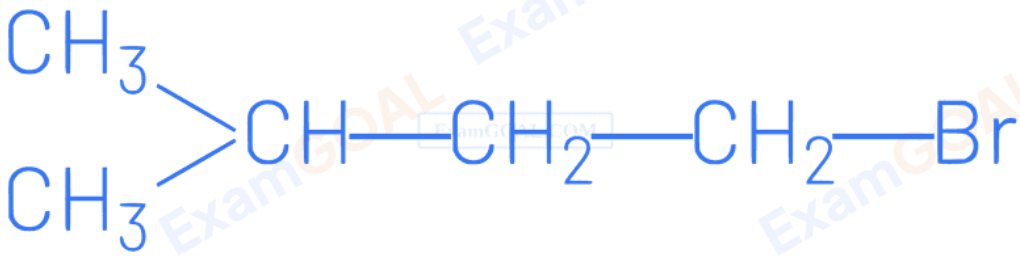
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Q12. NEET 2021

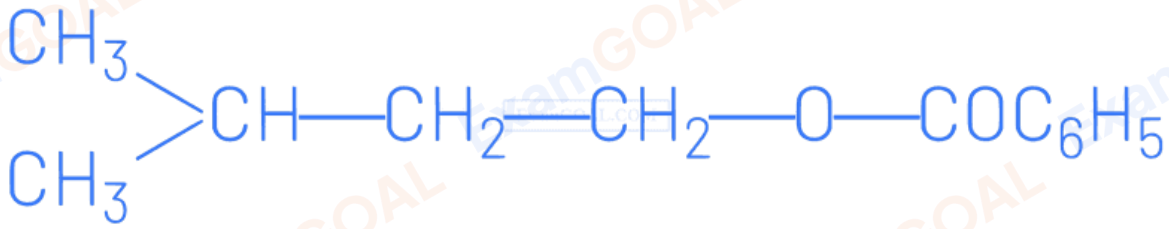
The major product of the following chemical reaction is :



(A)



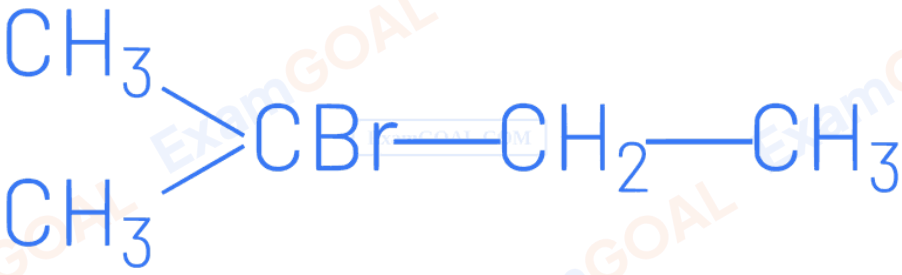
(B)



(C)



(D)



Open solution

Q13. NEET 2021

The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on?

(A) Huckel's Rule (B) Saytzeff's Rule (C) Hund's Rule (D) Hofmann Rule

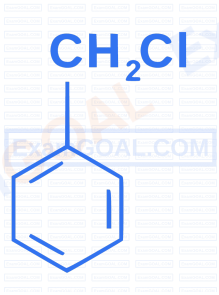
Open solution

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Q14. NEET 2020 Phase 2

Open solution

Which of the following will **NOT** undergo S_N1 reaction with $\overline{O}H$?

(A) 

(B) $CH_2 = CH - CH_2Cl$

(C) $(CH_3)_3CCl$

(D) 

Q15. NEET 2020 Phase 1

Open solution

Which of the following alkane cannot be made in good yield by Wurtz reaction?

(A) 2, 3-Dimethylbutane

(B) n-Heptane

(C) n-Butane

(D) n-Hexane

Q16. NEET 2020 Phase 1

Open solution

Elimination reaction of 2-Bromo-pentane to form pent-2-ene is :

(A) β-Elimination reaction

(B) Follows Zaitsev rule

(C) Dehydrohalogenation reaction

(D) Dehydration reaction

(A) (A), (C), (D)

(B) (B), (C), (D)

(C) (A), (B), (D)

(D) (A), (B), (C)

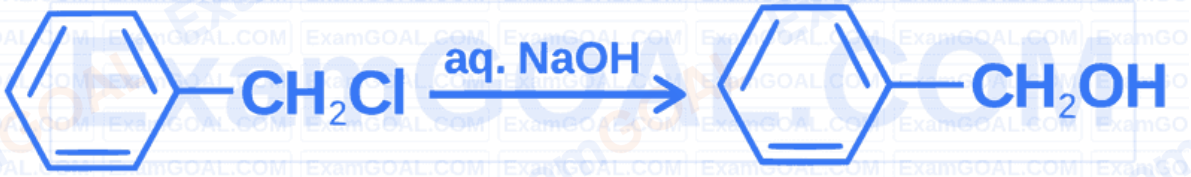
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Q17. NEET 2019 (Odisha)

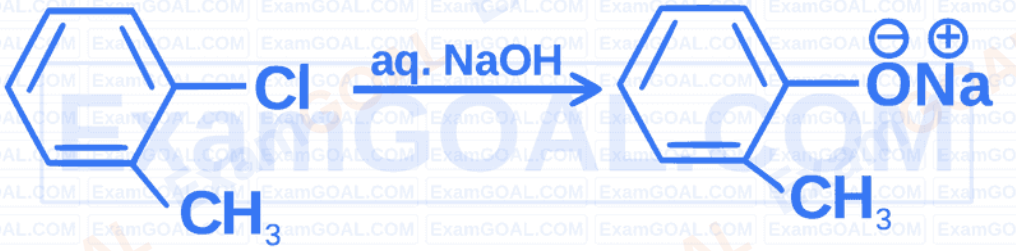
Open solution

The hydrolysis reaction that takes place at the slowest rate, among the following is

(A)



(B)



(C) $H_3C-CH_2-Cl \xrightarrow{\text{aq. NaOH}} H_3C-CH_2-OH$

(D) $H_2C=CH-CH_2Cl \xrightarrow{\text{aq. NaOH}} H_2C=CH-CH_2OH$

Q18. NEET 2018

Open solution

The compound C_7H_8 undergoes the following reactions :

$$C_7H_8 \xrightarrow{3Cl_2/\Delta} A \xrightarrow{Br_2/Fe} B \xrightarrow{Zn/HCl} C$$

The product 'C' is

(A) m–bromotoluene

(B) o–bromotoluene

(C) 3–bromo–2,4,6–trichlorotoluene

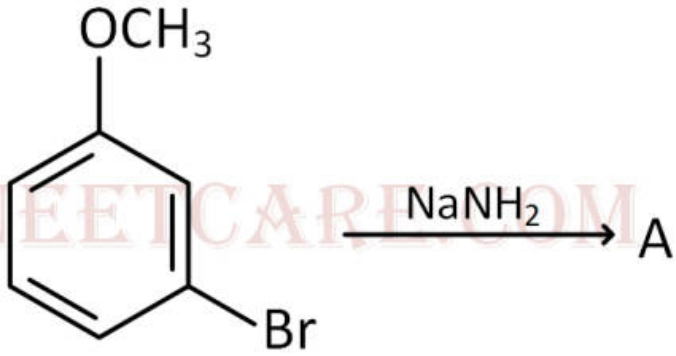
(D) p–bromotoluene

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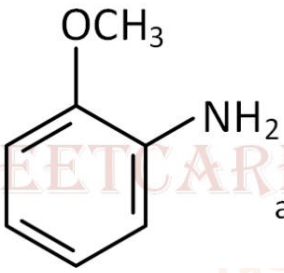
Q19. NEET 2017

Identify A and predict the type of reaction.

Open solution

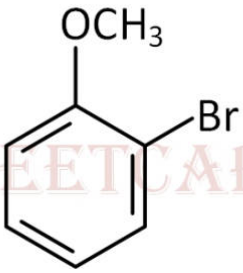


(A)



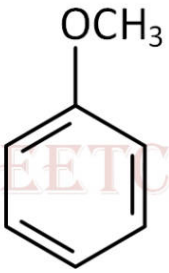
and elimination addition reaction

(B)



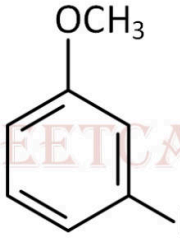
and cine substitution reaction

(C)



and cine substitution

(D)



and substitution reaction

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Q20. NEET 2016 Phase 2

Open solution

Consider the reaction,

$CH_3CH_2CH_2Br + NaCN \rightarrow CH_3CH_2CH_2CN + NaBr$

This reaction will be the fastest in

(A) ethanol

(B) methanol

(C) N, N' -dimethylformamide (DMF)

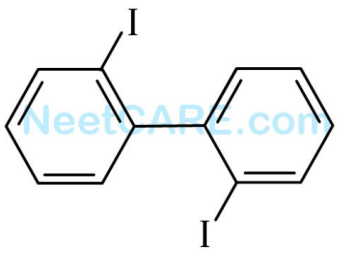
(D) Water

Q21. NEET 2016 Phase 1

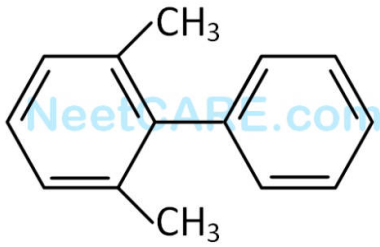
Open solution

Which of the following biphenyls is optically active ?

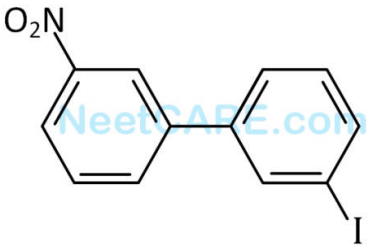
(A)



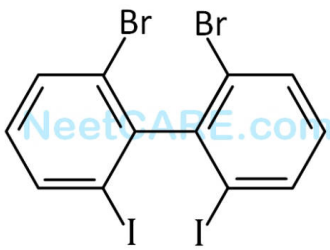
(B)



(C)



(D)



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Q22. AIPMT 2015

In an S_N1 reaction on chiral centers, there is

(A) inversion more than retention leading to partial racemisation

(B) 100% retention

(C) 100% inversion

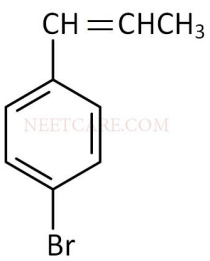
(D) 100% racemisation

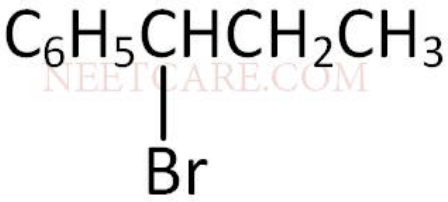
Open solution

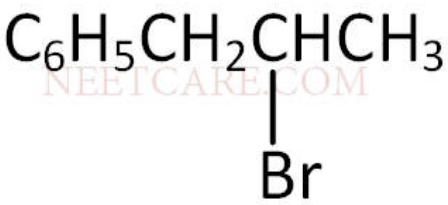
Q23. AIPMT 2015 Cancelled Paper

The reaction of $C_6H_5CH = CHCH_3$ with HBr produces

(A) $C_6H_5CH_2CH_2CH_2Br$

(B) 

(C) 

(D) 

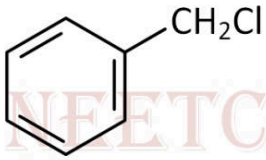
Open solution

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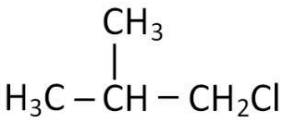
Q24. AIPMT 2014

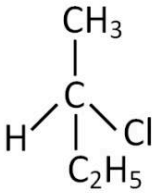
Which of the following compounds will undergo racemisation when solution of KOH hydrolyses?

Open solution

(i) 

(ii) $CH_3CH_2CH_2Cl$

(iii) 

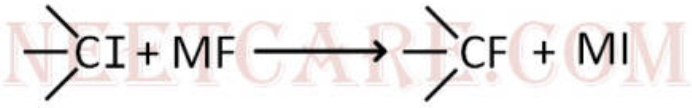
(iv) 

(A) (i) and (ii) (B) (ii) and (iv) (C) (iv) (D) (i) and (iv)

Q25. AIPMT 2012 Mains

In the replacement reaction

Open solution



The reaction will be most favourable if M happens to be

(A) Na (B) K (C) Rb (D) Li

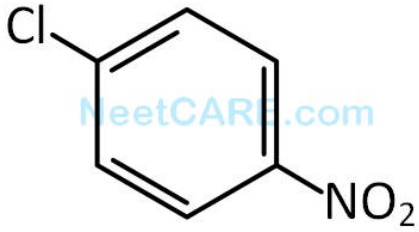
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Q26. AIPMT 2011 Mains

Open solution

Which of the following compounds undergoes nucleophilic substitution reaction most easily?

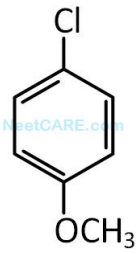
(A)



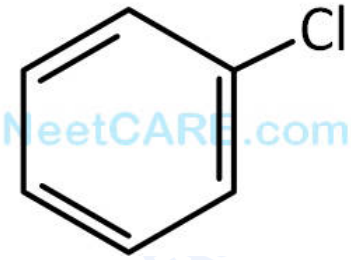
(B)



(C)



(D)



Q27. AIPMT 2011 Mains

Open solution

Consider the reactions.

(i) $(CH_3)_2CH - CH_2Br \xrightarrow{C_2H_5OH} (CH_3)_2CH - CH_2OC_2H_5 + HBr$

(ii) $(CH_3)_2CH - CH_2Br \xrightarrow{C_2H_5O^-} (CH_3)_2CH - CH_2OC_2H_5 + Br^-$

The mechanisms of reactions (i) and (ii) are respectively

(A) S_N1 and S_N2 (B) S_N1 and S_N1 (C) S_N2 and S_N2 (D) S_N2 and S_N1

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Q28. AIPMT 2010 Mains

Open solution

In the following reaction

$$C_6H_5CH_2Br \xrightarrow[2.\ H_3O^+]{1.\ Mg.\ Ether} X,$$

the product 'X' is

(A) $C_6H_5CH_2OCH_2C_6H_5$

(B) $C_6H_5CH_2OH$

(C) $C_6H_5CH_3$

(D) $C_6H_5CH_2CH_2C_6H_5$

Q29. AIPMT 2010 Prelims

Open solution

Which one is most reactive towards S_N1 reaction?

(A) $C_6H_5CH(C_6H_5)Br$

(B) $C_6H_5CH(CH_3)Br$

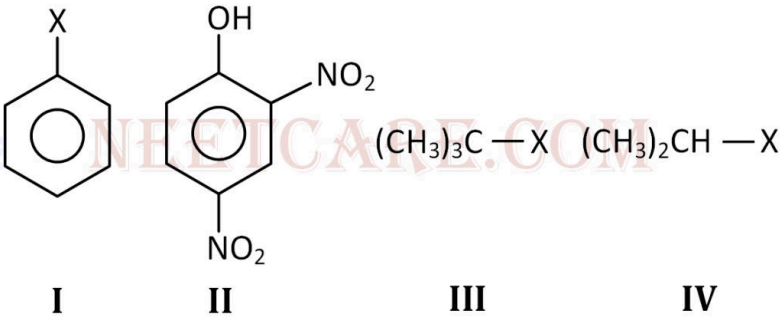
(C) $C_6H_5C(CH_3)(C_6H_5)Br$

(D) $C_6H_5CH_2Br$

Q30. AIPMT 2010 Prelims

Open solution

The correct order of increasing reactivity of



I II III IV

C—X bond towards nucleophile in the following compounds is

(A) I < II < IV < III

(B) II < III < I < IV

(C) IV < III < I < II

(D) III < II < I < IV

Q31. AIPMT 2009

Open solution

Which of the following reactions is an example of nucleophilic substitution reaction?

(A) $2RX + 2Na \rightarrow R - R + 2NaX$

(B) $RX + H_2 \rightarrow RH + HX$

(C) $RX + Mg \rightarrow RMgX$

(D) $RX + KOH \rightarrow ROH + KX$

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Q32. AIPMT 2008

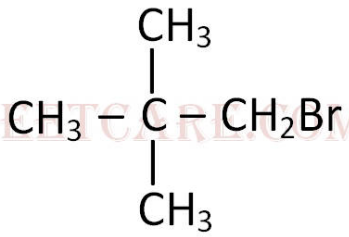
Open solution

In a S_N2 substitution reaction of the type

$R - Br + Cl^- \xrightarrow{DMF} R{-}Cl + Br^-$

which one of the following has the highest relative rate?

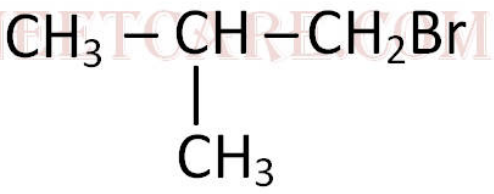
(A)



(B) CH_3CH_2Br

(C) $CH_3CH_2CH_2Br$

(D)



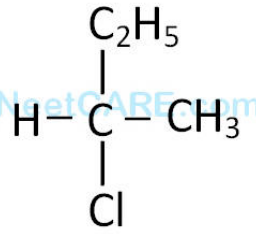
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Q33. AIPMT 2007

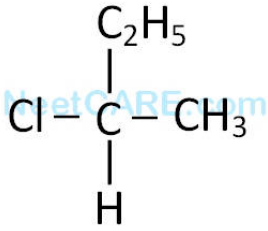
$CH_3 - CHCl - CH_2 - CH_3$ has a chiral centre. Which one of the following represents its R-configuration?

Open solution

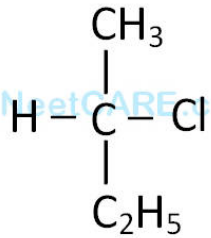
(A)



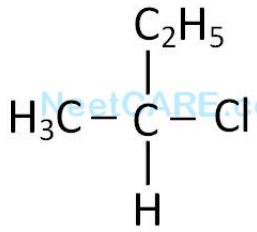
(B)



(C)



(D)



Q34. AIPMT 2006

Which of the following is not chiral?

(A) 2-Hydroxypropanoic acid
(B) 2-Butanol
(C) 2,3-Dibromopentane
(D) 3-Bromopentane

Open solution

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Q35. AIPMT 2005

Which of the following undergoes nucleophilic substitution exclusively by S_N1 mechanism?

(A) Ethyl chloride
(B) Isopropyl chloride
(C) Chlorobenzene
(D) Benzyl chloride

Open solution

Q36. AIPMT 2004

Which of the following is least reactive in a nucleophilic substitution reaction?

(A) $(CH_3)_3C-Cl$ (B) $CH_2{=}CHCl$ (C) CH_3CH_2Cl (D) $CH_2{=}CHCH_2Cl$

Open solution

Q37. AIPMT 2002

Reactivity order of halides for dehydrologenation is

(A) $R-F>R-Cl>R-Br>R-\mathrm{I}$
(B) $R-\mathrm{I}>R-Br>R-Cl>R-F$
(C) $R-\mathrm{I}>R-Cl>R-Br>R-F$
(D) $R-F>R-\mathrm{I}>R-Br>R-Cl$

Open solution

Q38. AIPMT 2002

$$CH_3CH_2Cl \xrightarrow{NaCN} X \xrightarrow{Ni/H_2} Y \xrightarrow{\text{acetic anhydride}} Z$$

NEETCARE.COM

Open solution

Z in the above reaction sequence is

(A) $CH_3CH_2CH_2NHCOCH_3$
(B) $CH_3CH_2CH_2NH_2$
(C) $CH_3CH_2CH_2CONHCH_3$
(D) $CH_3CH_2CH_2CONHCOCH_3$

Q39. AIPMT 2001

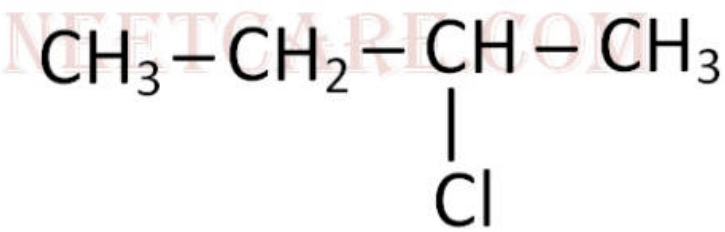
An organic compound A(C_4H_9Cl) on reaction with Na/diethyl ether gives a hydrocarbon which on monochlorination gives only one chloro derivative then, A is

(A) t-butyl chloride
(B) s-butyl chloride
(C) iso-butyl chloride
(D) n-butyl chloride

Open solution

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Q40. AIPMT 2001



Open solution

obtained by chlorination of n-butane will be

(A) meso form (B) racemic mixture (C) d-form (D) l-form

Q41. AIPMT 1998

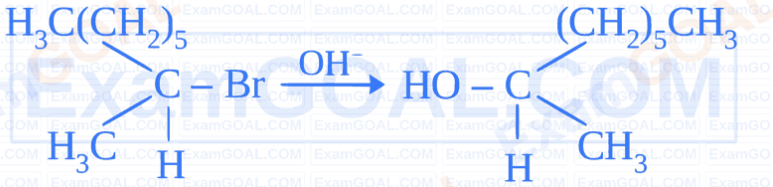
2-bromopentane is heated with potassium ethoxide in ethanol. The major product obtained is

(A) 2-ethoxypentane (B) pentene-1 (C) *trans*-pentene-2 (D) *cis*-pentene-2

Open solution

Q42. AIPMT 1997

The following reaction is described as



Open solution

(A) S_N2 (B) S_N0 (C) S_E2 (D) S_N1

Q43. AIPMT 1997

The alkyl halide is converted into an alcohol by

(A) elimination (B) dehydrohalogenation

(C) addition (D) substitution.

Open solution

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Q44. AIPMT 1997

Replacement of Cl of chlorobenzene to give phenol requires drastic conditions. But chlorine of 2,4-dinitrochlorobenzene is readily replaced because

(A) NO_2 donates e^- at *meta* position

(B) NO_2 withdraws e^- from *ortho/para* positions

(C) NO_2 make ring electron rich at *ortho* and *para*

(D) NO_2 withdraws e^- from *meta* position.

Open solution

Q45. AIPMT 1994

Grignard reagent is prepared by the reaction between

(A) magnesium and alkane

(B) magnesium and aromatic hydrocarbon

(C) zinc and alkyl halide

(D) magnesium and alkyl halide.

Open solution

Q46. AIPMT 1994

Reaction of *t*-butyl bromide with sodium methoxide produces

(A) sodium *t*-butoxide

(B) *t*-butyl methyl ether

(C) isobutane

(D) isobutylene.

Open solution

Q47. AIPMT 1993

When chlorine is passed through propene at 400°C, which of the following is formed?

(A) PVC

(B) Allyl chloride

(C) Propyl chloride

(D) 1, 2-Dichloroethane

Open solution

Q48. AIPMT 1993

Chlorobenzene reacts with Mg in dry ether to give a compound (A) which further reacts with ethanol to yield

(A) phenol

(B) benzene

(C) ethyl benzene

(D) phenyl ether.

Open solution

Q49. AIPMT 1993

Industrial preparation of chloroform employs acetone and

(A) phosgene

(B) calcium hypochlorite

(C) chlorine gas

(D) sodium chloride.

Open solution

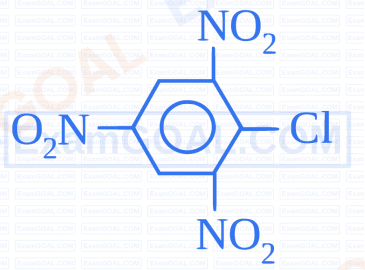
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Q50. AIPMT 1989

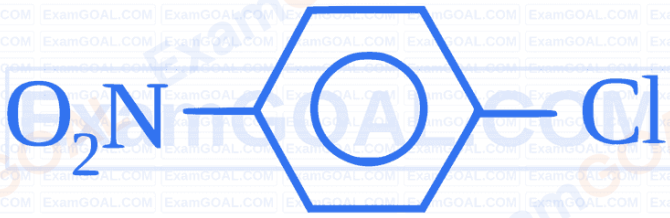
Which chloro derivative of benzene among the following would undergo hydrolysis most readily with aqueous sodium hydroxide to furnish the corresponding hydroxy derivative?

Open solution

(A)



(B)



(C)



(D) C_6H_5Cl

Q51. AIPMT 1988

Phosgene is a common name for

Open solution

(A) phosphoryl chloride
(B) thionyl chloride
(C) carbon dioxide and phosphine
(D) carbonyl chloride.

Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q52. AIPMT 1999 · MCQ (Single Correct Answer)

Gammexane is

Open solution

(A) chlorobenzene
(B) benzyl chloride
(C) bromobenzene
(D) benzene hexachloride

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Answer Key

MCQ (More than One Correct Answer)

1. **C**				

MCQ (Single Correct Answer)

2. **D**	3. **B**	4. **A**	5. **D**	6. **A**
7. **C**	8. **C**	9. **B**	10. **A**	11. **B**
12. **A**	13. **B**	14. **D**	15. **B**	16. **D**
17. **B**	18. **A**	19. **D**	20. **C**	21. **D**
22. **A**	23. **C**	24. **C**	25. **C**	26. **A**
27. **C**	28. **C**	29. **C**	30. **A**	31. **D**
32. **B**	33. **B**	34. **D**	35. **D**	36. **B**
37. **B**	38. **A**	39. **A**	40. **B**	41. **C**
42. **A**	43. **D**	44. **B**	45. **D**	46. **D**
47. **B**	48. **B**	49. **B**	50. **A**	51. **D**

Out of syllabus

52. **D**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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